

Question :

Task Description

Develop a "**Student Utility Program**" that performs the following operations using a menu-driven approach. Part A: Java Basics

- Display student information.
- Accept Name, Roll Number, and Marks.
- Calculate Total and Percentage.
- Display the result.

Part B: Conditional Statements

Implement menu options to:

- Check whether a number is Even or Odd.
- Find the Largest of Three Numbers.
- Display Grade based on percentage using if-else.
- Display Day of the Week using switch.

Part C: Looping Statements

Write programs to:

- Print multiplication tables.
- Display numbers from 1 to N.
- Calculate the sum of first N natural numbers.
- Generate the Fibonacci series.

Part D: Methods : Every method should be parameterized and should return the result:

Create user-defined methods for:

- Finding the factorial of a number.
- Checking whether a number is prime.
- Finding the maximum of two numbers.
- Calculating the area of a circle.

Call each method from the main() method.

Steps :

1. Steps of the Program

1. Import the Scanner class.
2. Create the StudentUtility class.
3. Create a Scanner object to take input.
4. Display the main menu.
5. Read the user's choice.
6. Use a do-while loop to repeatedly display the menu until the user exits.

Choice 1 – Student Information

7. Read the student's name and roll number.
8. Read marks in 3 subjects.
9. Calculate total marks.
10. Calculate percentage.
11. Display student details and result.

Choice 2 – Conditional Statements

12. Display the conditional statements menu.
13. **Even or Odd:** Check number % 2.
14. **Largest of Three:** Compare three numbers and display the largest.
15. **Grade:** Read percentage and assign a grade using if-else.
16. **Day of Week:** Read a number from 1–7 and display the corresponding day.

Choice 3 – Looping Statements

17. Display the looping statements menu.
18. **Multiplication Table:** Use a for loop to print the table.
19. **1 to N:** Use a loop to display numbers from 1 to N.
20. **Sum:** Use a loop to calculate the sum of the first N natural numbers.
21. **Fibonacci:** Use a loop to generate the Fibonacci series.

Choice 4 – Utility Methods

22. Display the methods menu.
23. **Factorial:** Call the factorial() method.
24. **Prime Check:** Call the isPrime() method.

25. **Maximum:** Call the `maxOfTwo()` method.
26. **Area of Circle:** Call the `areaOfCircle()` method.

Exit

27. If the user selects 0, display "**Exiting program...**".
 28. If an invalid choice is entered, display "**Invalid choice!**".
 29. Continue the menu until the choice is 0.
 30. Close the Scanner.
 31. **End the program.**
-

2. Overall Algorithm

1. **Start**
2. Create a Scanner object.
3. Repeat the following using a do-while loop:
 - Display the Student Utility menu.
 - Read choice.
4. If choice = 1:
 - Read student information.
 - Read 3 subject marks.
 - Calculate total and percentage.
 - Display the result.
5. If choice = 2:
 - Display conditional statement options.
 - Read `condChoice`.
 - Perform the selected operation:
 - Check even/odd.
 - Find largest of three numbers.
 - Determine grade.
 - Display day of the week.

6. If choice = 3:

- Display looping statement options.
- Read loopChoice.
- Perform the selected operation:
 - Print multiplication table.
 - Display numbers from 1 to N.
 - Calculate sum of first N natural numbers.
 - Generate Fibonacci series.

7. If choice = 4:

- Display utility method options.
- Read methodChoice.
- Perform the selected operation:
 - Calculate factorial.
 - Check prime number.
 - Find maximum of two numbers.
 - Calculate area of circle.

8. If choice = 0:

- Display "**Exiting program...**".

9. Otherwise:

- Display "**Invalid choice!**".

10. Repeat until choice = 0.

11. Close the Scanner.

12. **Stop.**

Code :

```
import java.util.Scanner;
```

```
public class StudentUtility {  
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);

int choice;

do {

    System.out.println("studentUtility program");

    System.out.println("1.Student Information");

    System.out.println("2.Conditional Statements");

    System.out.println("3.Looping Statements");

    System.out.println("4.Utility methods");

    System.out.println("5.Exit");

    System.out.println("Enter your choice:");

    choice=sc.nextInt();

    switch(choice) {

    case 1:

        System.out.println("Enter Name:");

        sc.nextLine();

        String name=sc.nextLine();

        System.out.println("Enter the Roll Number:");

        int roll =sc.nextInt();

        System.out.println("Enter Marks in 3 subjects");

        int m1=sc.nextInt(),m2=sc.nextInt(),m3=sc.nextInt();

        int total=m1+m2+m3;

        double percentage =(total/3)*100;

        System.out.println("\n--- Student Result ---");

        System.out.println("Name: " + name);

        System.out.println("Roll No: " + roll);

        System.out.println("Total Marks: " + total);

        System.out.println("Percentage: " + percentage + "%");

        break;
```

case 2:

```
System.out.println("\n Conditional Statement");  
System.out.println("1. Even or Odd");  
System.out.println("2. Largest of Three Numbers");  
System.out.println("3. Grade based on Percentage");  
System.out.println("4. Day of the Week");  
int condChoice = sc.nextInt();
```

```
switch (condChoice) {
```

case 1:

```
    System.out.print("Enter a number: ");  
    int num = sc.nextInt();  
    System.out.println(num % 2 == 0 ? "Even" : "Odd");  
    break;
```

case 2:

```
    System.out.print("Enter three numbers: ");  
    int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();  
    int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);  
    System.out.println("Largest: " + largest);  
    break;
```

case 3:

```
    System.out.print("Enter percentage: ");  
    double per = sc.nextDouble();  
    if (per >= 90) System.out.println("Grade: A+");  
    else if (per >= 75) System.out.println("Grade: A");  
    else if (per >= 60) System.out.println("Grade: B");  
    else if (per >= 40) System.out.println("Grade: C");
```

```

        else System.out.println("Grade: Fail");

        break;

    case 4:

        System.out.print("Enter day number (1-7): ");

        int day = sc.nextInt();

        switch (day) {

            case 1: System.out.println("Monday"); break;

            case 2: System.out.println("Tuesday"); break;

            case 3: System.out.println("Wednesday"); break;

            case 4: System.out.println("Thursday"); break;

            case 5: System.out.println("Friday"); break;

            case 6: System.out.println("Saturday"); break;

            case 7: System.out.println("Sunday"); break;

            default: System.out.println("Invalid day!");

        }

        break;

    }

    break;

case 3: // Part C

    System.out.println("\n--- Looping Statements ---");

    System.out.println("1. Multiplication Table");

    System.out.println("2. Display numbers 1 to N");

    System.out.println("3. Sum of first N natural numbers");

    System.out.println("4. Fibonacci Series");

    int loopChoice = sc.nextInt();

    switch (loopChoice) {

        case 1:

```

```
System.out.print("Enter number: ");  
  
int n = sc.nextInt();  
  
for (int i = 1; i <= 10; i++)  
  
    System.out.println(n + " x " + i + " = " + (n * i));  
  
break;
```

case 2:

```
System.out.print("Enter N: ");  
  
int N = sc.nextInt();  
  
for (int i = 1; i <= N; i++) System.out.print(i + " ");  
  
System.out.println();  
  
break;
```

case 3:

```
System.out.print("Enter N: ");  
  
int numN = sc.nextInt();  
  
int sum = 0;  
  
for (int i = 1; i <= numN; i++) sum += i;  
  
System.out.println("Sum = " + sum);  
  
break;
```

case 4:

```
System.out.print("Enter terms: ");  
  
int terms = sc.nextInt();  
  
int f1 = 0, f2 = 1;  
  
System.out.print("Fibonacci: " + f1 + " " + f2);  
  
for (int i = 3; i <= terms; i++) {  
  
    int next = f1 + f2;  
  
    System.out.print(" " + next);  
  
    f1 = f2;  
  
    f2 = next;
```



```
    }  
    System.out.println();  
    break;  
}  
break;
```

case 4: // Part D

```
System.out.println("\n--- Methods ---");  
System.out.println("1. Factorial");  
System.out.println("2. Prime Check");  
System.out.println("3. Max of Two Numbers");  
System.out.println("4. Area of Circle");  
int methodChoice = sc.nextInt();
```

```
switch (methodChoice) {
```

case 1:

```
    System.out.println("Enter number: ");  
    int factNum = sc.nextInt();  
    System.out.println("Factorial = " + factorial(factNum));  
    break;
```

case 2:

```
    System.out.println("Enter number: ");  
    int primeNum = sc.nextInt();  
  
    System.out.println(isPrime(primeNum) ? "Prime" : "Not Prime");  
    break;
```

case 3:

```

        System.out.print("Enter two numbers: ");

        int x = sc.nextInt(), y = sc.nextInt();

        System.out.println("Max = " + maxOfTwo(x, y));

        break;

    case 4:

        System.out.print("Enter radius: ");

        double r = sc.nextDouble();

        System.out.println("Area = " + areaOfCircle(r));

        break;

    }

        break;

    case 0:

        System.out.println("Exiting program...");

        break;

    default:

        System.out.println("Invalid choice!");

    }

} while (choice != 0);

sc.close();

}

public static long factorial(int n) {

    long fact = 1;

    for (int i = 1; i <= n; i++) {

        fact *= i;

    }

}

```

```
    return fact;
}
```

```
public static boolean isPrime(int n) {
    if (n <= 1)
        return false;

    for (int i = 2; i <= Math.sqrt(n); i++) {
        if (n % i == 0)
            return false;
    }
    return true;
}
```

```
public static int maxOfTwo(int a, int b) {
    return (a > b) ? a : b;
}
```

```
public static double areaOfCircle(double r) {
    return Math.PI * r * r;
}
}
```

Output :

```

studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
1
Enter Name:
abhinay
Enter the Roll Number:
264
Enter Marks in 3 subjects
90
80
90

--- Student Result ---
Name: abhinay
Roll No: 264
Total Marks: 260
Percentage: 86.67%
studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
2
Conditional Statement

```

```

Enter your choice:
4

--- Methods ---
1. Factorial
2. Prime Check
3. Max of Two Numbers
4. Area of Circle
4
Enter radius: 10
Area = 314.15926535897
studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:

```

```
2. Largest of Three Numbers
3. Grade based on Percentage
4. Day of the Week
1
Enter a number: 28
Even
studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
3

--- Looping Statements ---
1. Multiplication Table
2. Display numbers 1 to N
3. Sum of first N natural numbers
4. Fibonacci Series
1
Enter number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```